

ASTR 511 (O'Connell) Lecture Notes

TIPS FOR SUCCESS IN OBSERVATIONAL ASTRONOMY



Here are some tips for graduate students, which I think would be echoed by most of my colleagues, on how to approach research in observational astronomy.

A. Learn to ask the *right questions*

- In undergraduate school, you became good at answering questions posed by others. Now, you have to learn to **ask your own questions**. This is as important as any other aspect of front-line research.
- Learn to **rank** issues for overall importance to the field and for novelty.
- Here's a simple **thought experiment** you might try to put things in perspective:

Suppose you had access to a "*transient trans-temporal viewport*" that would allow you to read an elementary astronomy textbook from the year 2100 AD for one hour. What topics would you look up?

B. Learn to assess *solvability*

- Learn how to determine which questions/problems are ripe for progress.

- Solvability is a function both of the state of current knowledge and existing observational capabilities.
- Many "interesting" problems are **not** ripe for solution.
- Evaluate **signal-to-noise** issues; these are often critical to assessing solvability.
- Also assess the interests & strengths of **competing groups** & their potential for beating you to a result.

C. Develop a healthy *skeptical outlook*

- **Cultivated skepticism** is the cornerstone of science

"He believed in the primacy of doubt, not as a blemish upon our ability to know but as the essence of knowing."

----- J. Gleick, writing about physicist Richard Feynman.

- Apply to: established results, new results, and **(especially) your own results**

Be self-critical: make "reality check" your mantra.

Find your own mistakes before somebody else does.

D. Know the *scientific background* thoroughly

- Learn to **critically dissect the literature**, based on original, not secondary, sources and to integrate it into a coherent whole.
- For longer term projects (e.g. a thesis) it is important to learn the **entire history** of your field. You need to know how others got from "C" to "D" if you're hoping to go from "J" to "K." Among other benefits, there are many excellent insights that remain unexploited in the older literature.
- It is just as important to become familiar with the **wrong** ideas and **why** they were wrong as it is to know the currently accepted "right" ideas. There have been many more wrong ideas than right ones.
- Beware of the "emperor's new clothes." Always bring fresh eyes to any subject.
- Read the more important papers from hardcopy. **Annotate them or make written summaries** of key findings, strengths & weaknesses, things you don't understand. It helps to keep copies of related papers together in a binder.

NB: my preference for hardcopy is only partly because of old age. My definite impression is that using only screens leads to poorer comprehension and less careful writing (and thinking). It is often easier to cross-reference items in hardcopy. Controlled studies have also shown that taking handwritten rather than digital notes conveys better understanding. This phenomenon is not well understood, but it appears to be real.

Nonetheless, assistance in organizing digital copies of the more important literature is ever more valuable as the volume of published material continues to exponentiate (doubling time about 9 years). Using a reference management system like Mendeley is highly recommended for young scientists, though you can easily set up a [simple literature library](#) for yourself.

- In this area, your **memory** is one of your greatest personal assets. The ability to absorb, organize, correlate, synthesize, and recall large amounts of information is essential to a working scientist. Your K-12 teachers should have spent more time on that. Any fool can "look it up on the Internet."

E. Develop skill with *ROMPs* & *SWAGs*

- ROMPs & SWAGs = "rough order of magnitude problems" and "silly wild-ass guesses"
- Simple exploratory calculations are the **best way** to shape the early stages of any research program.
- The first three [ASTR 511 problem sets](#) contain good examples of ROMPs in basic observational astronomy.
- For an extended treatment, see, for example, S. Mahajan's MIT course on [Street Fighting Mathematics](#).

F. Track your own progress

- Set **milestones** and regularly **review** your progress towards them. Are you converging towards your goals?
- Keep a **regular journal** of your ideas (good & bad). If you're engaged with your subject, there should be entries almost every day.
 - You will find it essential to also keep detailed **logs/journals** concerning individual research projects. That includes tracking software development and journaling interactive computing sessions.
 - In some fields (e.g. biomedicine) it is common for people to keep notarized journals of daily work in order to establish priority for new discoveries and patentable ideas. Astronomers aren't as concerned with such issues, but it is certainly advisable to record potentially important insights.

G. Take quick advantage of *new observational capabilities*

- These are often the source of important new discoveries.
- A "new capability" can be as simple as large amounts of observing time on an existing facility or a small upgrade such as a new type of imaging filter or a higher dispersion grating in a spectrometer.
- You need a running start (and an informed assessment of the competitive climate) to

take advantage of early science in big new projects like JWST or LSST.

H. Learn to write good *observing proposals*

- Success rates for proposals on important facilities are small: typically in the range 10-40%.
- A compelling proposal must be clearly, persuasively, and concisely written and must demonstrate:
 1. That the questions you are asking are **important and relevant**;
 2. That the program is **technically feasible**;
 3. That you are **competent** to execute it; and
 4. That it will provide a **definitive answer** to the questions posed.
- You must write for harried, non-specialist TAC members who usually have only a **few minutes** to read each proposal and who are looking for **reasons to reject**
- For details on planning and writing proposals in this competitive environment, see my ["Tips on Writing Proposals in Astronomy"](#).
- **Practice** writing proposals in conjunction with your advisors before you leave the shelter of graduate school.

I. Understand your *instruments* well

- This will take a minimum of 2-3 observing runs with a given instrument.
- Learn their limits and how to push the envelope.

J. Understand *analysis techniques* well

- It is doubtful that your early reductions/analysis for a given problem will be acceptable in the long run. Plan for **iterations**, possibly extensive.
- **Test** techniques on synthetic data sets where you know the answer.
- You will need to develop a good working familiarity with **statistical methods** of analyzing observations and hypothesis testing, an increasingly sophisticated element of astronomical programs.

K. Dealing with *computers*

- Computers are **tools, not science**.
- Despite appearances, computers have **not** increased the capacity of the human brain to absorb and process information. You will have to work hard to distill and clarify what computers are telling you.
- Don't trust and **always verify**. You can't really do this unless you understand clearly how

computers work, and this means that...

You will almost certainly have to become a capable computer **programmer**, not merely a user of pre-packaged software. It's useful to know how to code in more than one language, assuming the need arises naturally out of the science. The most popular computer operating system in astronomy at present is UNIX/LINUX/macOS, and the most popular programming language is Python.

The advent of petabyte-scale data-driven astronomy in the 2020's makes coding experience yet more valuable.

The rapid growth of applications of **artificial intelligence** in astronomy poses a major quandary for users. Even if you write your own AI programs, the details of their actual operation may remain opaque. Although they make dealing with huge datasets more tractable, assessing their biases and reliability often requires very extensive testing. But even that may not suffice, as amply demonstrated by the spectacular failures of various commercial AI products, which behave more like digital slot machines than reliable research tools.

- Grad school in astronomy provides excellent exposure to **high-end computing** in the form of sophisticated applications and simulations programming and large data sets. If this becomes your main interest and you decide on a career outside of academic science that emphasizes computing, your best approach is to obtain your astronomy degrees as quickly as possible, learning more about computing in your spare time, and then focus all your energies on the new field. Astronomy degrees are good credentials in the computing technology sector.
- Social media? For pre-professionals, involvement with social media should be cautious. Some of it is useful, much is wasted time, a big share is pernicious and imposes high social and professional costs. It can become an unproductive time-sink, and the visibility and permanence of unfiltered thoughts have cratered many a career. **Beware.**

L. Dealing with advisors

- Most advice comes from long experience; **don't disregard it lightly**. In troublesome areas, seek advice from several senior people.
- Look at any research project or task assigned you, no matter how outwardly routine, as an **opportunity** to grow as a scientist beyond the nominal boundaries. You may well have new insights or see new avenues to exploit that your overloaded advisor has missed or ignored.
- In the course of a PhD project, you are expected to become **independent** of your advisor and at least as knowledgeable as s/he is on that subject. You should know what is the "**next step**" before being told.

M. Dealing with groups

A glance at the ApJ will demonstrate the growing **dominance of group science**. The

rise of groups is a natural consequence of the increasing scope and complexity of modern astronomical research. New instruments are almost always shepherded by groups. Group work is also an increasingly important mechanism for marshalling diverse existing resources to do large-scale science (e.g. the supernova surveys, the deep cosmic fields, gravitational waves, SDSS, LSST, etc).

Being a member of a large group provides a rapid education in how professional astronomy is conducted and also cushions your initiation because substantial resources and expertise are there to help you. Large groups, especially international ones, allow you to quickly establish a **network** of scientists who become familiar with you and your work.

On the other hand, you have to work harder to establish a perspective and reputation **independent** of the group.

Successful professional groups are based on **complementary strengths, not mutual weaknesses** (unlike the group learning that is now fashionable in educational circles). Therefore:

- Develop a unique expertise, skill
- Produce; be responsible
- Think for yourself...but learn how to interact with other members cordially and constructively
- Lead by example
- Be sure you develop a reputation/expertise that **distinguishes** you from other group members. In practice that implies, at a minimum, being lead author on some of the group's publications.

N. Publishing

- A good goal by PhD time is to have published in the major journals **one paper for every year you've been in graduate school**. You should be lead author of some of these.
- **Learn to write good, clear, concise scientific prose**. Most entering graduate students cannot do this.
 - Consult style guides for tips, e.g. the *Guide to Science Writing* from the Journal of Young Investigators and *The Elements of Style* by Strunk & White.
 - The more practice, the better. Write up for yourself brief reviews of the literature or summaries of your interim results in full journal style. These serve as good writing practice, as drafts for polished versions or presentations, and to place your accomplishments in the context of the main issues you are trying to address. Ask other students to read and critique your work.

- Learn to write well **before you begin to submit work to your advisors**. You want them spending time on your ideas, not grammar, spelling, or word usage.
- Be aware of the **“reader pyramid”**: few people will read the whole paper; more will read the introduction and conclusion; many more will skim/read the abstract.
 - Therefore, be sure the abstract contains all the key points, and be sure the intro/conclusion are clear and complete. The conclusion should emphasize what is **new or unique** about the work.
- Set high standards for yourself (and your co-authors) in writing. Reputations are based on the number of **good** papers one produces, not the total number of papers. It is not always fair, but “quality” is often judged by the number of **citations** a paper receives (which is easily tracked by various literature servers).

O. Presentations

- Learn to give interesting and effective presentations on all time scales from 5 minutes to an hour. Analyze and emulate others who do this well.
- **Real-time rehearsals are the only way** to prepare well for your early ventures in this arena.

An essential part of rehearsals is to **formulate answers** to all the likely questions, especially skeptical ones, from your audience. Try to view your presentation as an outsider would.

Good rehearsals are the best way to quell stage-fright, which everybody has to some extent.

- Unfortunately, you will need to learn PowerPoint or the equivalent. Just bear in mind that PowerPoint is the messenger, not the message. Compare A/V presentations by others, and base your approach on the most effective of those. You can capture copies of speakers' screen displays during Zoom sessions.
- See [these tips on making presentations](#) from UVa ASTR 8500 students.

P. Teaching

- The best way to understand a subject is to try to **teach it** to somebody else. Volunteer to give research talks, tutorials, paper reviews, etc., to your colleagues.
- Universities now expect viable faculty job candidates to have some **experience with teaching**, preferably undergraduate and in the form of complete responsibility for development and delivery of at least one course. At UVa your best opportunity for that is to teach in the summer session. Teaching experience is valuable for a professional career whether or not you are aiming for a job in higher education.
- Take any teaching assignment whether as TA or instructor seriously and plan to do a

good job. Among other things, that means learning how to handle **two disparate responsibilities** (teaching and studying/research) simultaneously.

- If you are interested in what awaits you in teaching large undergraduate courses, see my [lecture](#) or [article](#) on *Facts of Life for New Teachers in the Astronomy Non-Majors Curriculum*.
- **Mentoring** younger people is, of course, an important part of any professional career. Whether or not you choose to participate as a mentor during graduate school, you should take advantage of whatever formal training or informal advice you can find.

Q. Outreach

- Outreach to the general public has always been embraced at some level by most astronomical institutions (in distinction to most other sciences). It has lately assumed greater importance, such that applicants for academic jobs are expected to have **some experience or interest** here.
- Outreach can take many forms. Telescope viewing is most common. But other varieties include observatory tours, public lectures, astronomy clubs, K-12 tutorials and activities, K-12 teacher training, planetarium and museum displays, press liaisons and spokespersons, and so forth. Online versions include course webpages, information sites, blogs, videos, and podcasts, among others. You have many options, but it is best to choose areas where there is good departmental experience and support. Your outreach activities will have to be fitted around your principal academic responsibilities.
- An ever more prominent form of outreach is the self-generated press release that describes ostensibly important research discoveries. Large group science coupled with regular funding cycles breed institutional imperatives for publicizing results. Consult the history of "big physics" (particle accelerators, etc) for the sociology here. The deluge of press releases causes the media to mistake incremental results for fundamental ones.

You might ponder the wisdom of influential physicist Richard Garwin here:

*"In life you either get things done or get credit for things getting done.
Take your pick."*

R. Scientific Discovery, Scientific Careers

The list of tips above constitutes tactics; but what about **strategy**? What is the best path to scientific discovery or a good scientific career?

Discovery first.

Broadly, there are two kinds of discovery: recognition of something new or a definitive interpretation of known phenomena. The former is easier for young people. For the latter, you usually need greater exposure to the field. Although many "interpretational" discoveries are theoretical, others are observational (e.g. Hubble's discovery of Cepheid variables in M31, which instantly resolved the "island

universe" controversy; or the identification of gamma-ray bursts with distant galaxies).

Scientific discoveries emerge from some combination of "the prepared mind," resources, opportunity, and, inevitably, luck. There is probably about equal weight to those four components, and there's no way to successfully engineer them. But follow the chain in order. The better prepared you are---the more you know and have produced---the more likely it is that the good resources you seek will be available to you. Opportunity may follow. You have to wait for luck. Whatever form that takes, it's essential that you be able to recognize a favorable coincidence of opportunity and luck, which means that you must actively cultivate an **alertness** for them.

Discovery and innovation are rarely associated with very young scientists. All the low-hanging fruit was harvested long ago. You have a foundational mountain to climb before you can approach new territory. In pop-psychology this is sometimes referred to as the "10,000 hours" standard: it probably takes 10,000 hours of close engagement with the field to become ready for real contributions. The clock starts running probably in your junior year of college. **You can't "think out of the box" until you know what's inside it.**

Scientific discovery also requires **creativity and imagination**. Those are terms that people tend to think are confined to areas like literature, art, music, or film. But scientists are compelled to develop an intuition for complex phenomena so far removed from everyday experience that active imaginations are essential to make progress. Mathematics is critical here, both to facilitate and to discipline imagination. Scientific creativity is needed to see beyond the data, to sense the underlying reality and connectivity, and to build a coherent conceptual framework involving all the related phenomena. This kind of creativity can be much more demanding than, say, writing a novel or making a film.

You obviously can't discover something if you aren't looking, so discovery depends also on **effort and persistence**. Noticing something that is unexpected is often an entree to discovery, but this means you must know what is expected.

The essence of a successful scientist is **curiosity**.

Careers? You need a **plan**, and you need to think actively about it. Your plan should build on your interests and inclinations, but it must be shaped by the **realities**, economic and otherwise, of the field. You will require a number of **skills** which you have only in embryonic form now, and part of your plan must be to act deliberately to develop them further.

Brainpower in professional astronomy counts --- but less than it did in your undergraduate career because (almost) everybody in the field is bright and capable. Instead, **self-motivation, hard work, persistence, focus, efficiency, and good time-management are critical to success.**

Training in most graduate programs is **"T-shaped"**. You are expected to become

acquainted with the basics of many subfields of astronomy (the crossbar) while acquiring deep knowledge in at least one (the upright). The whole "T" is important. The narrow/deep component is essential to understanding how scientific research actually progresses. But from a career standpoint, you must also develop a broad understanding of the field and versatile skills that are transferable to other research areas. You will likely also find insights from other subfields valuable in your own.

"The optimum strategy for a scientist is to know something about everything and everything about something." --- Vitaly Ginzburg

Your immediate aim by PhD time should be to become one of the **leading authorities** on some area of significant current interest. It is expected that this area will usually be of modest scope, but when conference organizers are picking the most knowledgeable younger speakers for review talks, you want your name to be on the short list.

This means that you must not only have important expertise but that **others must know you have it**. That's the essence of **networking**, which is ever more important for young scientists. The more you talk science with your graduate student colleagues, local scientists, and outsiders, the better. The more you publish, the better. Attend topic-oriented meetings as well as the large annual AAS meetings and **make as many contacts** as practicable. Keep in mind that the future leaders of the field are among your contemporaries.

It's useful to have a **very brief memorized summary** -- perhaps 30 seconds -- of your current interests (an "elevator pitch") at your disposal whenever you will encounter people you might want to impress.

You will make many **mistakes** during graduate school. But everybody does. Don't let them discourage you -- in fact, you should welcome learning about your shortcomings in the protected environment of graduate school, when you have the time to address them. Good scientists learn from their mistakes and disappointments and incorporate them into a renewed attack on the issue.

Resilience is an essential attribute of successful scientists.

Research topics? Paradoxically, it is not necessarily best to get into the currently "hot" subject areas. Those may be where the money is and where your mentors are; but these areas tend to overproduce PhDs, and there will be tough competition from experienced scientists. Ideally, you want to be at the leading edge of a **new** wave of research that peaks about **ten years from now**. But, obviously, it's not easy to figure out what that might be. Some "hot" areas will persist, but others won't. Seek advice on prospects for the next decade before deciding on your primary research program.

No matter how promising the field you choose to work in, **keep developing those transferable skills and interests; keep looking around the corner**.

Web Links

For more resources regarding careers, see [UVa ASTR 8500](#), where the topic is professional development for astronomy graduate students.



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